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in francesco-dondi
📷 Fdondi

Born 29/10/1990
Italian citizen, C permit
Married, no children



Francesco Dondi

SOFTWARE ENGINEER - BACKEND & SYSTEMS

Who I am

Ex-Google Software Engineer with strong foundations in algorithms, distributed systems, and performance optimization. Proven ability to design, build, and scale production systems, with a background in mathematics and competitive programming. Experienced across backend engineering, data infrastructure, and ML systems, with a focus on solving complex technical problems efficiently.

2025-current: Consultant with Randstad

- 09/2025 - 11/2025 **Software Engineer (Contract), Randstad Digital/YouTube**
- Upgraded complex legacy data routing systems in YouTube backend
 - Debugged and resolved production issues impacting data flow and system reliability
 - Improved maintainability of legacy systems through refactoring

2023-2025: AI startups era

- 03/2025 - 05/2025 **Backend Engineer, Hoshii**
- Designed and extended backend APIs (Go) supporting a production webshop
 - Integrated frontend (React) with backend services and MongoDB
 - Improved internal AI orchestration pipelines for better performance and reliability
- 09/2024 - 01/2025 **Senior Machine Learning Engineer, Giotto AI**
- Worked on ARC challenge research with focus on data augmentation strategies
 - Built scalable multi-GPU training pipelines using PyTorch Lightning
 - Applied transformer models (BART) for classification tasks
- 03/2024 - 05/2024 **Senior Software Engineer, Deep Impact**
- Developed computer vision systems across cloud and embedded environments (Nvidia Jetson)
 - Improved containerization and deployment via Docker
 - Diagnosed and fixed system-level bugs across heterogeneous environments

2021-2023: Cloud Data Warehousing arc

- 03/2023 - 09/2023 **Senior Software Engineer, DoubleCloud**
- Built infrastructure around ClickHouse-based data systems
 - Automated deployments using Terraform, Helm, and SaltStack
 - Developed plugins in Go for observability tooling (Telegraf)
- 06/2021 - 09/2022 **Senior Software Engineer, Firebolt**
- Contributed to SQL query optimization engine for high-performance analytics database
 - Improved test infrastructure to support large-scale validation
 - Enhanced SQL parsing and optimization capabilities

2020-2021: Finance interlude

- 08/2020 - 02/2021 **Software Developer / Quantitative Analyst, F-Trust**
- Developed and maintained trading engine in C++
 - Designed and migrated data storage solutions (PostgreSQL)

Core Skills

Programming

C++
Python
Go



Systems

Backend Engineering
Distributed Systems
Optimization



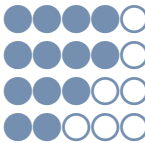
Fundamentals

Data Structures
Complexity Analysis
Problem Solving



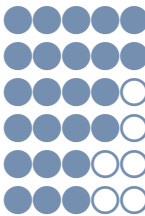
Infrastructure

CI/CD
Linux
Docker
Terraform



Data & ML

SQL
- Clickhouse SQL
- PostgreSQL
PyTorch
TensorFlow
Data Pipelines



Languages

English
Italiano
Deutsch
Français



2017-2020: Google

- 10/2017 - 05/2020 **Software Engineer, Google (Search/Knowledge Graph)**
- Productionized experimental internal tool into scalable system used in production
 - Reduced runtime from days to hours through algorithmic and systems optimizations
 - Designed features for handling new data types and improving system flexibility
 - Improved observability, monitoring, and usability for internal users

2015-2017: Initial Experiences

- 04/2016 - 10/2017 **Software Developer, Ascent Software**
- Developed embedded C++ drivers for automotive communication
 - Worked on real-time systems (QNX OS)
- 04/2015 - 04/2016 **Junior Developer, Rulx Inc./CNR**
- Optimized algorithms using parallel computing (OpenMP, CUDA)
 - Migrated MATLAB algorithms to Python

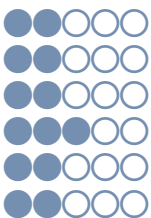
Formal education

- 02/24-11/24 **CAS ETH in Machine Learning in Finance and Insurance, ETH Zürich**
- 2009-2015 **Galileian School of Higher Education, University of Padua**, Enriching program, 98/100
- 2012-2014 **Master Degree in Mathematics, University of Padua**, 108/110
- 2009-2012 **Bachelor Degree in Mathematics, University of Padua**, 110/110 cum Laude

Projects

- Aug 2025 **Azure Affirmation website** dondi.fyi/affirmation
- Azure Static Web App with Azure Function App backend, which offers works of affirmation by either sampling them from a Cosmos MongoDB, or generating through OpenAI API.
- Aug 2024 **AI Keyboard** github.com/Fdondi/ai_keyboard
- A keyboard to perform various operations with LLMs with one click. Working on Bluetooth connection.

Azure
Azure Functions
Cosmos DB
C++ Embedded
Arduino
Electronics



Hackatons

- 14-16/Nov/2025 **HackaHealth Zürich, ETH Student House**
Help Patrick paraglide again despite paralysed legs.
- 26-27/Sep/2025 **ZurichHack, Swiss {ai} Weeks**
Mine the Gap: Detect anomalies in an evolving knowledge graph.
- 11-12/Sep/2025 **Energy Data Hackdays, BKW**
Smart Design of Resilient Local Energy Communities
- 19-20/Mar/2016 **H-Farm, H-Ack Food**, Reinventing the 'Bio food' experience.
12-13/Dec/2015 **H-Farm, H-Ack Luxottica**, Ideas for the future of glasses.

Prizes

- 2024 **Codility Pi Challenge**, Silver Award
- 2009 **Italian Olympics of Mathematics**, Gold 🏆
- 2008 **Italian Olympics of Information Technology**, Bronze 🏆

KEY STRENGTHS FOR GOOGLE

- Strong algorithmic foundation (mathematics + competitive programming)
- Proven experience in production systems and performance optimization
- Prior Google experience with demonstrated impact and positive feedback from colleagues
- Ability to debug complex systems across multiple layers
- Versatility across backend, infrastructure, and data systems

Leaderboards/training

📖 Leetcode

📖 Kaggle

Books to swear by

📖 **Designing Data-Intensive Applications** Martin Kleppmann

📖 **Kubernetes Patterns**
Bilgin Ibryam, Roland Huß

📖 **Effective Java** Joshua Bloch

Personal

Hobbies: Read, hike, bike, swim.